

Theory of Computation – PYQ and Suggestions

1. What is a finite automaton? Explain the difference between DFA and NFA.

Answer:

- A **finite automaton (FA)** is a mathematical model of computation used to recognize **regular languages**.
 - **DFA (Deterministic FA)**: Each state has exactly one transition for each symbol.
 - **NFA (Nondeterministic FA)**: A state can have zero, one, or more transitions for a symbol; can include ϵ -transitions.
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2. Explain the concept of ϵ -closure in NFA.

Answer:

- The **ϵ -closure** of a state is the set of states reachable from it by **ϵ -transitions only**.
 - Used in converting **NFA to DFA**.
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3. Define Pumping Lemma for Regular Languages. Give its significance.

Answer:

- **Pumping Lemma**: If L is regular, there exists an integer p such that any string $s \in L$ with $|s| \geq p$ can be divided as $s = xyz$, satisfying:
 1. $|y| > 0$
 2. $|xy| \leq p$
 3. For all $i \geq 0$, $xy^iz \in L$
 - **Significance**: Used to **prove that a language is not regular**.
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4. What is a context-free grammar (CFG)? Write its components.

Answer:

- A CFG is a 4-tuple **$G = (V, \Sigma, R, S)$** :
 1. **V** = set of variables (non-terminals)
 2. **Σ** = set of terminals
 3. **R** = set of production rules ($A \rightarrow \alpha$, $A \in V$, $\alpha \in (V \cup \Sigma)^*$)
 4. **S** = start symbol
 - CFGs generate **context-free languages (CFLs)**.
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5. Explain ambiguity in CFG. Give an example.

Answer:

- A CFG is **ambiguous** if there exists a string with **more than one leftmost or rightmost derivation**.

- **Example:**

Grammar: $E \rightarrow E+E \mid EE \mid id$

String: *id+idid* has **two different parse trees**.

6. Define derivation tree (parse tree).

Answer:

- A **derivation tree** is a tree that represents the application of CFG rules to generate a string.
 - Root = Start symbol, leaves = terminals of the string.
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7. Explain Chomsky hierarchy of languages.

Answer:

Type	Grammar	Language	Machine
0	Unrestricted	Recursively enumerable	Turing machine
1	Context-sensitive	CSL	Linear bounded automaton (LBA)
2	Context-free	CFL	Pushdown automaton (PDA)
3	Regular	RL	Finite automaton (DFA/NFA)

8. What are closure properties of Regular Languages?

Answer:

- Regular languages are closed under:

1. Union: $L_1 \cup L_2$
2. Intersection: $L_1 \cap L_2$
3. Complement: L_1'
4. Concatenation: $L_1 L_2$
5. Kleene Star: L_1^*
6. Reversal

9. Define linear bounded automaton (LBA).

Answer:

- An LBA is a **Turing machine with restricted tape**.
- Tape size is bounded by the **input length**.
- Recognizes **context-sensitive languages (CSL)**.

10. Why is Type-1 grammar called context-sensitive?

Answer:

- In Type-1 grammar, productions are of the form $\alpha A \beta \rightarrow \alpha \gamma \beta$, where $\gamma \neq \epsilon$.
- A non-terminal **A** can be replaced depending on the surrounding context **(α, β)**, hence **context-sensitive**.

11. Define Turing machine and its components.

Answer:

- A Turing machine (TM) is a **mathematical model of computation** capable of simulating any algorithm.
- **Components:** ($Q, \Sigma, \Gamma, \delta, q_0, q_{\text{accept}}, q_{\text{reject}}$)
 1. **Q** = set of states
 2. **Σ** = input alphabet
 3. **Γ** = tape alphabet ($\Sigma \subseteq \Gamma$)
 4. **δ** = transition function
 5. **q_0** = start state
 6. **$q_{\text{accept}}, q_{\text{reject}}$** = accept/reject states

12. Define recursively enumerable (RE) and recursive languages.

Answer:

- **Recursively enumerable (RE):** TM may halt on strings in L, may not halt otherwise.

- **Recursive:** TM halts for every input, accepts strings in L, rejects strings not in L.

13. Define left recursion in CFG and how it is removed.

Answer:

- **Left recursion:** $A \rightarrow A\alpha \mid \beta$, leads to infinite recursion in top-down parsing.
- **Removed by:**
 1. Introduce new non-terminal A' .
 2. Rewrite:

$$A \rightarrow \beta A'$$

$$A' \rightarrow \alpha A' \mid \epsilon$$

14. Explain right-linear and left-linear grammars.

Answer:

- **Right-linear grammar:** $A \rightarrow aB$ or $A \rightarrow a$
- **Left-linear grammar:** $A \rightarrow Ba$ or $A \rightarrow a$
- Both generate **regular languages**, equivalent to **finite automata**.

15. Define language. How many languages are possible for an alphabet $\Sigma = \{a, b\}$?

Answer:

- **Language:** Any subset of Σ^* (set of all strings over Σ).
- $\Sigma = \{a, b\}$, strings = $\{\epsilon, a, b, aa, ab, ba, bb, \dots\}$
- Number of languages = $2^{(\text{number of strings in } \Sigma^*)}$, which is infinite.

16. Prove that regular languages are closed under intersection.

Answer (Idea):

- Let L_1 and L_2 be regular, with DFA $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$, $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$
- Construct DFA $M = (Q_1 \times Q_2, \Sigma, \delta, (q_1, q_2), F_1 \times F_2)$
- $\delta((p, q), a) = (\delta_1(p, a), \delta_2(q, a))$
- M accepts $L_1 \cap L_2$, so regular languages are closed under intersection.

17. Define Myhill-Nerode theorem (short).

Answer:

- It provides a **necessary and sufficient condition for regularity:**

- A language L is regular $\Leftrightarrow$ it has **finite equivalence classes** under relation: $x \sim y$ iff $\forall z, xz \in L \Leftrightarrow yz \in L$.
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18. Define pushdown automaton (PDA) and its relation to CFG.

Answer:

- PDA is a finite automaton with a **stack**, used to recognize **context-free languages (CFL)**.
 - **Relation:** For every CFG, there exists an equivalent PDA, and vice versa.
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19. Explain non-determinism in Turing machines.

Answer:

- **Non-deterministic TM (NTM):** Can move to **multiple states for the same input and tape symbol**.
 - Accepts a string if **any computation path leads to accept**.
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20. What is a regular expression? How is it related to FA?

Answer:

- **Regular expression (RE):** Symbolic representation of **regular languages** using operations: union, concatenation, Kleene star.
- **Equivalence:** Every RE $\leftrightarrow$ Finite Automaton.